3 A spring is used to join two frictionless trolleys A and B of mass M_1 and M_2 respectively, where $M_1 > M_2$ as shown in Fig. 3.1.

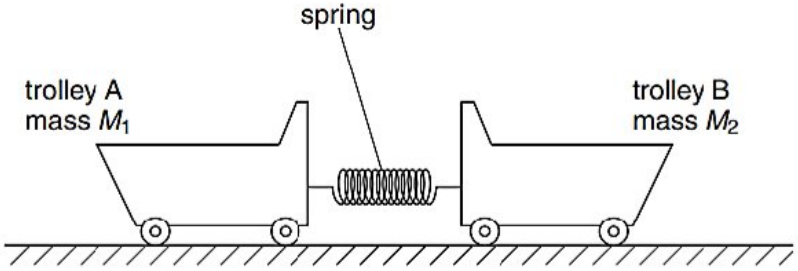


Fig. 3.1

The trolleys rest on a horizontal frictionless surface and are held apart so that the spring is extended.

The trolleys are then released.

(a) During the motion, the magnitude of the force exerted by the spring on each trolley varies with time.

With reference to Newton's laws of motion, explain why the magnitude of the impulse delivered to trolley A is equal to that delivered to trolley B.

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..... [3]

(b) Explain why the centre of gravity of the two-trolley system remains at rest throughout the motion.

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..... [2]

(c) Determine an expression for the ratio of the kinetic energy of trolley A to that of trolley B in terms of the masses of the two trolleys.

[3]